Candidate's Assessment Number.....

THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM II PRE-MOCK ASSESSMENT

031 **PHYSICS**

Time: 2:00 Hours **March, 2025**

Instructions

1. This paper consists of section A, B and C with a total of ten (10) questions.
2. Answer all questions.
3. All writing must be in blue or black ink except drawings which must be in pencil.
4. Calculators and communication devices are not allowed in Examination.
5. Write your Examination Number at the top on every page.
6. Where necessary the following constants may be used;
   (i) Acceleration due to gravity g = $10m/s^2$
   (ii) Density of water =$1g/cm^3$ or $1000kg/m^3$

FOR EXAMINERS USE ONLY		
QUESTION'S NUMBER	**SCORE**	**EXAMINER'S INITIALS**
1		
2		
3		
4		
5		
6		
7		
8		
9		

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10		
TOTAL		

SECTION A (15 marks)

1. For each of the following (i) – (x), choose the correct answer for the given alternative and write its letter in the box provided.

(i) A physics teacher instructed you to measure the volume of a stone with irregular shape. Which of the following pairs of instruments will you recommendto use?

A. metre rule and beaker
B. measuring cylinder and overflow can
C. measuring cylinder and a metre rule
D. metre rule and overflow can

(ii) Which statement tells the truth about the laboratory rules?

A. Making students enjoy science subject
B. Making students conduct experiment freely.
C. Ensuring safety while in the laboratory
D. Ensuring good communication with other students and the teacher

(iii) A student observed a colour changes when a tea bag is dipped into a cup containing hot water. Identify the phenomenon on this observation.

A. surface tension B. capillarity C. Diffusion D. Osmosis

(iv) Which of the following statements explain the meaning of upthrust?

A. Weight of body in air B. Actual weight of a body
C. Weight of liquid displaced D. Apparent weight.

(v) A rectangular block of density 0.8g/cm³ has dimension $0.5m \times 0.8m \times 6m$. What is the maximum pressure would it exert on the ground in N/m^2?

A.4800 B.480 C.48 D.4000

(vi) Which of the simple machines is a second-classlever?

A. Scissors B. Wheelbarrow C. Pliers D. Shovel

(vii) A man weighing 45kg climbs up a stair case to a height of 7 m in 5seconds. What is the amount of work done by a man?

A.300J B.3500J C.3150J D.3015J

(viii) What is a name representing a point at which resultant magnetic flux density is zero?

A.Neutral point B. North point C. South point D. magnetic field point

(ix) A boy come with a material which allow small portion of light to pass throughbut cannot see through it. What is the name of this material as classified in terms of light?

A. Transparent B.Translucent C.Luminous D. Opaque

(x) A cylinder of mass 25g when empty and volume 3000cm^{3}is full of liquid **Q.**The mass of the cylinder when full of liquid **Q** is 120g. What is the density of liquid **Q?**

A. 0.003g/cm. B.0.3g/cm³ C.0.03g/cm³ D. 3g/cm³

Answers For Question 1

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)

2. Match each item in List A with corresponding correct response in List B by writing the letter of the correct response below the number of the corresponding item in List A in the table provided.

LIST A	LIST B
(i) Protect buildings and other structures during thunderstorm. (ii) Produce charge on uncharged body without actual touching. (iii) Detecting the sign of the charge on a body. (iv) Store charges for later use. (v) Flows to the ground or to the object during earthing.	A. Charging by friction B. Electrons C. Lightning D. Capacitor E. Lightning conductor F. Gold leaf electroscope G. Charging by induction

Answers For Question 2

(i)	(ii)	(iii)	(iv)	(v)

SECTION B (70 marks)

3. (a) State the law of conservation of energy

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……

(b) Describe the energy changes which occur in each of the following processes:

(i) When water falls from a high point to the ground

.....

(ii) Using solar panel to light a bulb.

.....

4. (a) State two conditions for a body to be in equilibrium.

(i) ……

(ii) ……

(b) Figure 1, shows two boys carrying a ladder of weight 800N. Calculate the force P and Q that each boy support.

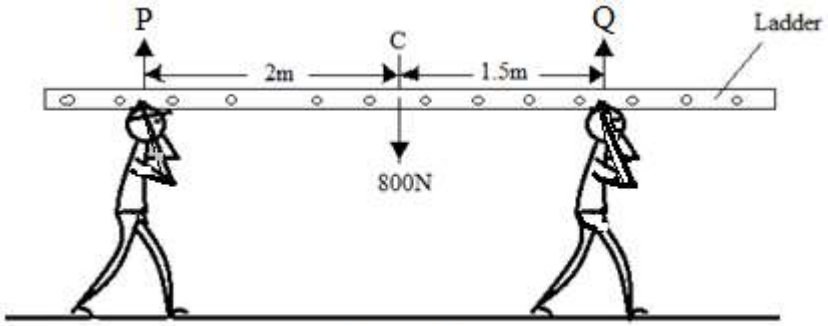


Figure 1

5. (a) Explain why are dams constructed thicker at the bottom than at the top?

.....……

……

(b) A human of mass 64kg is standing on sand soil with heels shoes of area $2cm^2$

(i) Find the pressure exerted by the human on the ground

(ii) Why does her heel sink into the ground?

……

……

6. (a) Distinguish between the following physical quantities as applied in motion in a straight line:

(i) speed and velocity

……

……

(ii) Distance and displacement

……

……

(b) A girl runs twice round a field track of length 500m

(i) What distance does she covered?

(ii) What is her displacement from the starting point?

7. (a) The image can be formed by a pin hole camera and a plane mirror. State any two differences of the images formed in those two cases

(i) .....

(ii) .....

(b) (i) A car advertisement claims that headlights of a new car can see around corners. Why is this statement untrue? Give onereason.

.....

.....

(ii) Figure 2, shows a ray of light PQ striking a mirror AB. The mirror AB and the mirror CD are at the right angles to each other QN is the normal line to the mirrorAB. If the object is placed at P, how many images will be formed?

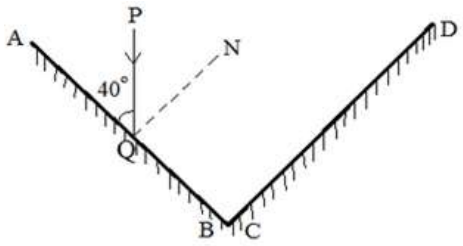


Figure 2

8. (a) A physics teacher instructed you to design a capacitor using two parallel plates. Explain any two factors that will you consider in order to design a capacitor of high value of capacitance.

(i) ……

(ii) ……

(b) What value of capacitor could be used to replace a set of 5μF, 10μF and 15μF capacitors connected in series.

9. (a) The north pole N of a magnet is stroked along a metal bar in the direction as shown in figure 3.

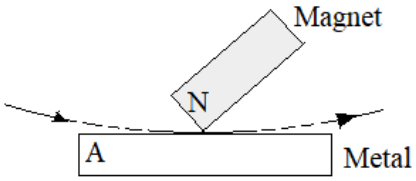


Figure 3

(i) Name a metal which would become permanently magnetized by stroking in this way. ……

(ii) If such a metal were used, what pole would be produced at A?

……

(b) States any three applications of magnets

(i) ……

(ii) ……

(iii) ……

SECTION C (15 Marks)

10.(a) Give the meaning of the following terms

(i) Electric circuit

……

……

(ii) Electric current

.....……

……

(b) How current electricity is useful in our daily life? Give two points

(i) .....

(ii) .....

(c) You are provided with a battery, a switch, a bulb and wires.

(i) Draw a simple electric circuit in which the bulb will operate.

(ii) How will you connect the ammeter with the bulb in order to measure the current flowing through it?

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